

JOSE: Reconstructing Privileged State from Joint History for Humanoid Motion Control

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Abstract—Deploying a humanoid policy trained on privileged simulator states means inferring those states on hardware, where joint encoders and an IMU are the usual sensors. The IMU carries base orientation and angular velocity, but its signal drifts and is disturbed by foot-strike impacts. During stance, leg kinematics constrain base motion relative to the supporting feet, motivating learned prediction of inertial state from joint history. We therefore test whether encoder history alone suffices to recover a teacher’s required states, the inertial components among them, without losing control performance. We present JOSE (Joint-Only State Estimator), a recurrent estimator that maps a window of joint-encoder history to the teacher’s privileged state, the orientation and angular velocity normally read from an IMU included, and hands it to a frozen teacher that acts with no IMU in the control loop. Training uses simulator-state regression with dataset aggregation. On three motion-tracking teachers trained with Adversarial Motion Priors and one command-conditioned locomotion teacher, driving the frozen teacher with JOSE halves the motion error of joint-only action distillation and tracks locomotion commands nearly as well as ground-truth observations. Reconstructing the inertial state rather than reading it leaves control unaffected by IMU noise, though encoder noise still degrades it without noise-aware training.

Index Terms—whole-body control, legged locomotion, humanoid robots, state estimation, proprioception

I. INTRODUCTION

A common approach to deploying a policy trained with privileged simulator information is to transfer its behavior to a student using observations available on the robot [1], [2]. Such students can encode information about unmeasured states implicitly, without explicitly reconstructing those states. Joint encoders and an IMU commonly provide proprioceptive inputs, although the observation sets and estimation pipelines differ across methods [2]–[5]. This reliance on inertial sensing raises the question of whether joint history alone can provide sufficient information for control.

Contact-aided estimators exploit leg kinematics to constrain base motion relative to the supporting feet [6], [7]. These constraints motivate testing whether joint history can predict the teacher’s required inertial states within the evaluated motion and terrain distributions. We therefore ask whether a privileged teacher can retain its control performance when its IMU-derived inputs are replaced by predictions from joint history.

If joint history can support this replacement, removing the IMU could also offer practical benefits at deployment. It would eliminate a sensor, along with its cost, failure modes, and calibration requirements. It would also remove reliance on inertial measurements that can be disturbed by locomotion

impacts (e.g., foot-ground impacts that introduce noise into accelerometer readings) [8].

Two routes reach this encoder-only setting. One distills the teacher into a student that maps encoder history straight to actions, without an IMU. The other predicts the privileged state explicitly and lets the frozen teacher act on it, reusing the teacher’s control policy rather than learning a new one from action-matching alone. We take the second route and compare against the first throughout. To this end we introduce JOSE (Joint-Only State Estimator), a recurrent estimator that recovers the teacher’s privileged state from a window of joint-encoder history, driving a frozen teacher from joint encoders alone with no IMU in the loop (Fig. 2). Because these strategies can reach similar survival rates yet still produce different motions, we also report motion fidelity relative to the teacher. Our teachers are trained with Adversarial Motion Priors (AMP) [9] for walking, dancing, and jumping, and with PPO [10] for command-conditioned locomotion. This work makes four contributions:

- We introduce JOSE, a recurrent estimator that reconstructs a frozen teacher’s privileged state, including the base orientation and angular velocity normally read from an IMU, from joint-encoder history alone.
- We show that comparable survival rates can mask substantial differences in motion fidelity, and that explicit state reconstruction better preserves the teacher’s motion than joint-only action distillation.
- We ablate the estimator over its architecture, history length, dataset aggregation, and observed joint set to select its best form.
- We evaluate JOSE against state-estimation and distillation baselines on motion fidelity and command tracking, including yaw-rate tracking, and characterize its robustness to sensor noise, terrain variation, and external pushes.

II. RELATED WORK

A. Teacher-Student Distillation and State Estimation

Teacher-student action distillation combines privileged learning [11] with output imitation [12]. In humanoid whole-body control, OmniH2O [13] and ExBody2 [14] train a deployable student $\pi_S(h)$ to imitate a privileged teacher $\pi_T(o^d, o^p)$ from observation history. The student maps this history straight to actions, so the teacher’s privileged state is captured implicitly in the learned policy rather than reconstructed.

Input estimation instead supplies a control policy with inferred state or latent variables. Rapid Motor Adaptation

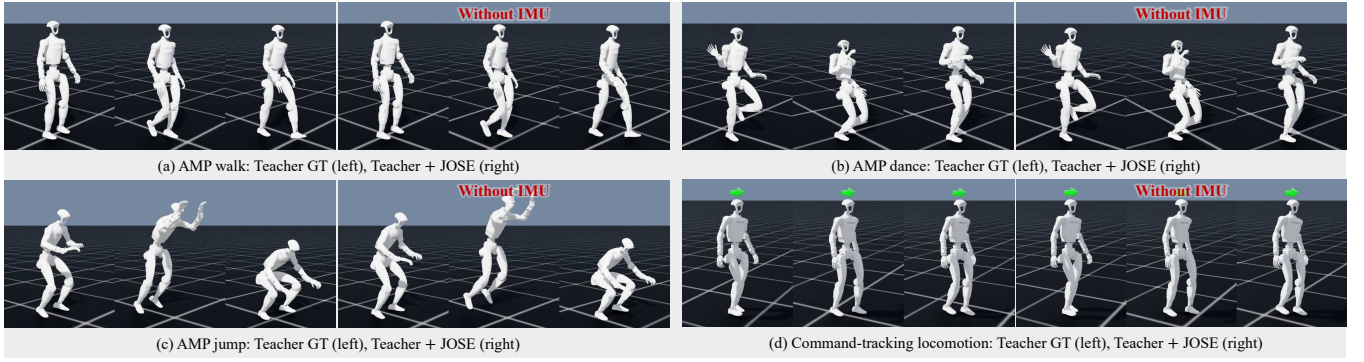


Fig. 1. **Frozen teacher policies driven by JOSE.** For each motion shown in (a)–(d), the left strip shows the teacher acting on ground-truth privileged observations, and the right strip shows the same teacher with those observations replaced by JOSE’s estimate from joint-encoder history, read without an IMU. The two strips stay close throughout, so the teacher’s motion is preserved from encoders alone.

(RMA) [3] predicts a latent encoding of environmental factors, such as friction and payload, from state and action history and supplies it to the base policy, and its bipedal extension fine-tunes the policy to absorb imperfect estimates [15]. Closer to our setting, the State Estimation Transformer (SET) [4] predicts explicit base height and linear velocity, and concurrent estimators recover base velocity and foot contact [5]. However, these approaches retain IMU-derived inputs, although the specific observation sets differ across methods.

JOSE differs on both the target and the interface. It reconstructs the privileged state, including base orientation and angular velocity commonly supplied by IMU-based sensing, from joint-encoder history alone, and it feeds a frozen teacher, so the teacher’s state-to-action map is retained rather than relearned.

B. Inertial-Kinematic State Estimation

Classical estimators fuse inertial measurements with leg kinematics and contact constraints. Examples include an extended Kalman filter for humanoid state estimation [7], a contact-aided invariant filter with improved convergence and consistency properties [6], and contact-centric leg odometry that uses joint velocities and contact-dependent updates to reduce drift [8]. GAIT extends this sensor-fusion approach by combining learned velocity and uncertainty estimates from inertial-leg attention with an invariant Kalman filter [16]. These methods demonstrate the value of kinematic constraints for state estimation while retaining inertial sensing. JOSE instead tests learned reconstruction from joint history without IMU inputs.

C. Covariate Shift and Fidelity to a Reference

Dataset aggregation (DAgger) addresses the distribution shift between an expert’s trajectories and the states a learned policy visits under its own errors [17]. Although it is normally applied to action imitation, JOSE meets the same shift while regressing states, since it is trained on the teacher’s states yet queried on the states its own estimates induce. We therefore apply it to state regression, labeling the visited rollouts with simulator ground truth.

Motion fidelity complements survival in character animation and humanoid control [9], [18]. Following human pose estimation [19], humanoid controllers and teleoperation systems report global and root-relative joint-position errors [13], [20], [21]. We use these metrics to compare each deployed controller with its teacher.

III. METHODOLOGY

A. Problem Formulation

We split the teacher’s observation into inputs available under encoder-only deployment, o_t^d , and the privileged states o_t^p (Fig. 2).

$$o_t = [o_t^d, o_t^p], \quad o_t^d \in \mathbb{R}^{2n+\alpha}, \quad o_t^p \in \mathbb{R}^{d_p}. \quad (1)$$

The available inputs o_t^d hold the positions and velocities of the $n = 29$ actuated joints, along with α further terms available at deployment but not read from the encoders, such as previous actions. This α is not fixed and varies with the teacher. These are all known at deployment and pass straight to the teacher, so none is reconstructed. JOSE reads the $2n$ encoder channels and reconstructs the privileged state o_t^p from them. The target o_t^p holds the remaining state a teacher requires, namely base orientation and angular velocity together with base linear velocity, height, or key body positions depending on the teacher. These targets are clean simulator states, including the inertial quantities an IMU would otherwise measure. Their contents and dimension d_p are given in Section III-B.

The teacher $\pi_T(o_t^d, o_t^p)$ is trained with the full observation. At deployment we drop o_t^p and replace it with an estimate produced by JOSE from a window of encoder history,

$$\begin{aligned} \hat{o}_t^p &= E_\theta(h_t), & a_t &= \pi_T(o_t^d, \hat{o}_t^p), \\ h_t &= (e_{t-W+1}, \dots, e_t), \end{aligned} \quad (2)$$

where e_t is the $2n$ -dimensional encoder reading at step t . We train one estimator for each of the four teachers (Fig. 1(a)–(d)) and keep the teachers frozen throughout estimator training and deployment.

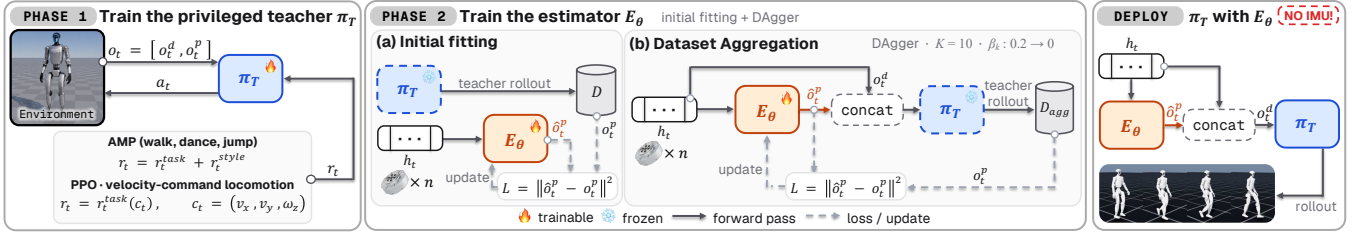


Fig. 2. **Training and deployment with JOSE.** Phase 1 trains a teacher π_T on the complete observation $o_t = [o_t^d, o_t^p]$, where o_t^d is the joint-encoder reading and o_t^p the privileged state. Phase 2 freezes π_T and trains JOSE to predict o_t^p from a window of encoder history h_t , first (a) on teacher rollouts and then (b) by iterative DAGger against simulator ground truth. Deployment feeds $[o_t^d, \delta_t^p]$ to the frozen teacher from joint encoders alone, with δ_t^p supplied by JOSE in place of the IMU.

B. Teacher Policies

We train four privileged teachers that differ in objective and in what they require of o_t^p . Three track reference motions and one follows a velocity command. JOSE reads the same encoder window in every case, and only the contents and dimension d_p of its prediction target change with the teacher.

The three motion-tracking teachers are trained with AMP [9] on a walking, a whole-body dance, and a jump clip from LAFAN1, retargeted to the G1 [22] and ordered from the near-periodic walk to the ballistic jump with a flight phase. Their reward is the AMP style term matching the clip, with no task term, and only walk adds a forward-velocity term. Reproducing a reference motion requires the full base state, namely base height, base orientation as its tangent and normal axes [23], base linear velocity, base angular velocity, and ten key body positions relative to the base, giving $d_p = 1 + 6 + 3 + 3 + 30 = 43$.

The fourth teacher is command-conditioned and trained with PPO [10] using velocity-tracking rewards, without an AMP style reward. It tracks a velocity command $c_t = (v_x, v_y, \omega_z)$ resampled periodically. Its PPO hyperparameters, scene, action space, and terminations follow uni-tree-rl_lab [24], while its reward set and command sampler are adapted from Isaac Lab’s G1 flat-terrain example [25]. Without a reference pose to reproduce, this teacher needs fewer reconstructed components than the motion-tracking teachers, namely base linear velocity, base angular velocity, and the gravity vector projected into the base frame, giving $d_p = 3 + 3 + 3 = 9$.

C. Recurrent Estimator

JOSE, denoted E_θ , is a two-layer LSTM with 256 hidden units that maps a window of $W = 25$ encoder frames to o_t^p , reading all 29 joints. It minimizes mean squared error, with inputs and targets standardized to zero mean and unit variance. It is fit for 50 epochs on the initial teacher-driven dataset and 10 epochs per DAGger round, using Adam at 10^{-3} , halved during aggregation, with a batch of 1024. Physics steps at 200 Hz and the policy acts every fourth step, for a 50 Hz control rate.

We vary the initial conditions across training episodes while holding the environment fixed. For motion tracking, each episode starts at a random phase of the reference clip. For locomotion, we randomize the initial base position within

± 0.5 m in x and y , yaw over the full circle, and joint velocities within ± 1 rad/s.

D. Dataset Aggregation

After an initial fit on teacher rollouts (Fig. 2(a)), we apply DAGger [17] to collect the states visited under estimated inputs (Fig. 2(b)). At round k , ground-truth inputs control a fraction β_k of environments and the remainder run on estimates. The encoder histories and their simulator-state targets are added to the dataset before retraining. Over $K = 10$ rounds, β_k decreases linearly from 0.2 to 0, so control shifts onto the estimator as it improves. Each round collects 2000 steps across 256 environments, and the aggregated set is capped at 250k samples by uniform random subsampling, so later rounds re-weight the distribution rather than grow it.

E. Deployment

At each control step, JOSE updates its encoder window and supplies the estimated state to the teacher. Estimation starts immediately, with the unavailable history initialized to zero. Beyond the joint encoders, no IMU or foot-contact measurement is used. The command-conditioned teacher additionally stacks a five-step history of its own observation terms as its PPO input, and for the reconstructed terms this stack is filled with JOSE’s estimates. This teacher-side history is separate from JOSE’s $W = 25$ encoder window, which is unchanged.

IV. EXPERIMENTS & RESULTS

We first ablate JOSE to select its configuration, comparing estimator architectures, history lengths, aggregation rounds, and observed joint sets. We then compare the selected configuration against baselines on survival, motion fidelity, and command tracking, and test its robustness against the same baselines under sensor degradation, terrain variation, and turning.

A. Setup

Everything runs in Isaac Lab [25] on a Unitree G1 at a 50 Hz control rate. Episodes are capped at 20 s for walk and locomotion and 10 s for dance and jump, and each closed-loop evaluation runs 200 episodes over three training seeds.

Locomotion commands. Survival and episode length on the locomotion teacher are measured under commands from the task sampler, with $v_x \in [0, 1]$ m/s, $v_y \in [-0.5, 0.5]$ m/s, and $\omega_z \in [-1, 1]$ rad/s sampled uniformly every 10 s and 2%

of environments commanded to stand. Command tracking instead uses fifteen fixed commands spanning zero and the axis limits, each evaluated for 4 s after a 1 s settling window, with the same commands for every method.

Comparison baselines. We compare JOSE against three baselines, and in the control tables JOSE and SET represent the frozen teacher driven by each estimator, while Teacher uses ground-truth observations. *Distillation* is the standard deployable-student recipe, a student imitating the frozen teacher’s actions from a history of joint encoders and an IMU while the teacher labels the states it visits, as in OmniH2O [13]. *Joint-only distillation* is the same student and budget with the six IMU-derived channels removed, isolating whether encoder history alone robustly reproduces the teacher’s behavior. *SET* [4] is a learned state estimator that regresses base height and linear velocity from proprioception and reads orientation and angular velocity from the IMU, so it represents the standard practice of reconstructing only the unmeasured states while trusting the IMU for the rest.

Metrics. Episodes terminate when the base height falls below a task-specific threshold or when the time limit is reached, and survival is the fraction reaching the time limit. Motion fidelity to the teacher is the global mean per-joint position error $E_{g\text{-mpipe}}$ and root-relative error E_{mpipe} , in millimetres, between the deployed controller and the teacher on ground-truth observations, rolled out from the same seeded initial condition over 100 control steps. A frozen teacher driven by an estimate does not trivially match the teacher, since a poor estimate diverges under the closed-loop dynamics, so a low error means the estimate keeps the loop near the teacher’s trajectory. $E_{g\text{-mpipe}}$ includes translation and heading drift, while E_{mpipe} removes root translation but retains heading. For locomotion, we report command-tracking RMSE separately for forward velocity, lateral velocity, and yaw rate. The normalized tracking error averages the three axis RMSEs after division by their respective fixed scales of 1.0 m/s, 0.5 m/s, and 1.0 rad/s. We use this dimensionless score to compare methods within each task. Δ teacher denotes the difference between a method’s normalized tracking error and that of the ground-truth-driven teacher. We report reconstruction RMSE between the estimated privileged state $\hat{o}_t^p$ and the corresponding simulator ground truth during closed-loop rollouts driven by the evaluated estimator. Reconstruction RMSE is the square root of the mean squared error over either the components of each state quantity or all target components, with the latter used only for within-task comparisons. In Table II this is reported per quantity, with Ori. the six tangent and normal orientation components and g the projected gravity, both dimensionless, height and bodies in metres, linear velocity in m/s, and angular velocity in rad/s. Tables report mean $\pm$ standard deviation across seeds, and figure error bars span the observed range.

B. Ablations

Four ablations select JOSE’s configuration, each varying one design choice while holding the others fixed, namely

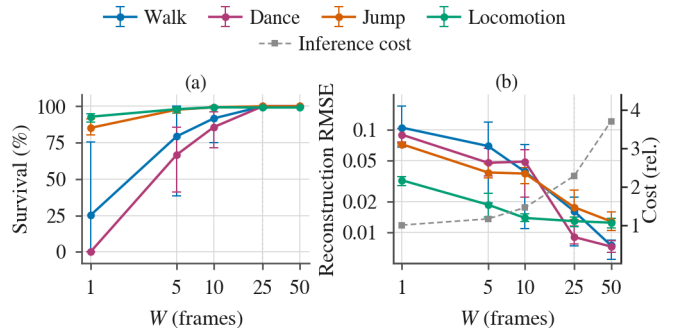


Fig. 3. **Window length ablation study.** Longer histories improve survival rate and reconstruction but increase inference cost, so we use $W = 25$.

the estimator architecture, the observed history length, the number of aggregation rounds, and the observed joint set.

1) *Estimator Architecture:* Table II compares the LSTM against a TCN [26] and a history MLP, matched in head and training, differing only in how they consume the window. The LSTM leads on control and reconstruction alike, with the lowest error on every state quantity and a dance linear-velocity RMSE more than an order of magnitude below both alternatives (0.0082 against 0.1232 and 0.1326 m/s). The alternatives collapse on the harder motions, the TCN failing every walk and dance episode and the history MLP every dance episode, so we use the LSTM throughout.

2) *Window Length:* Figure 3 compares survival (Fig. 3(a)) against reconstruction RMSE and relative inference cost (Fig. 3(b)) across history lengths, where inference cost is the compute per forward pass relative to the $W = 25$ estimator. A longer window reconstructs states better yet tracks the teacher worse in the loop, since extending to $W = 50$ lowers estimation error at $1.6\times$ the cost but raises global position error on walk (16.2 against 9.1 mm) and dance (12.9 against 5.9 mm). Since survival already saturates at $W = 25$ and lower regression error does not carry over to closed-loop fidelity, we use $W = 25$.

3) *DAGger Rounds:* Figure 4 compares aggregation rounds. At $K = 0$, training uses only teacher-driven data and every seed fails on all four tasks (Fig. 4(a)), which shows that reconstruction alone, without closed-loop data, does not sustain control. Survival then rises over the rounds, jump saturating first and walk last, while locomotion reconstruction RMSE keeps decreasing through $K = 10$ (Fig. 4(b)).

Walk is non-monotonic. Survival falls from 65–86% across seeds at $K = 4$ to zero at $K = 5$, then recovers to 98% at $K = 6$, so for each task we deploy the round with the best closed-loop performance rather than the final one.

4) *Observed Joint Set:* Model-based estimators use leg kinematics together with inertial measurements to constrain base state [6], [7]. This motivates testing how much of JOSE’s reconstruction performance can be retained using leg-joint history alone. Whole-body motions such as dance add upper-body coupling, however, so we test it by training JOSE on all joints, the legs, or the upper body, holding the estimator and aggregation fixed. Leg joints alone yield fidelity and survival close to all joints, both reaching 100%

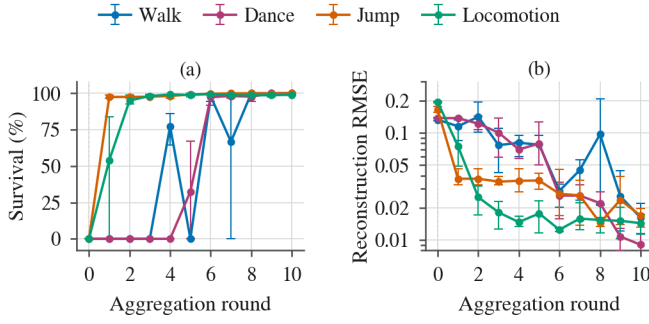


Fig. 4. **Aggregation rounds ablation study.** Closed-loop updates over DAGger rounds generally improve survival rate and reconstruction.

TABLE I
OBSERVED JOINT SET ABLATION.

Task	Joints	Surv. (%) $\uparrow$	Ep. len. $\uparrow$	$E_{g\text{-mpjpe}} \downarrow$	$E_{mpjpe} \downarrow$	Recon. RMSE $\downarrow$
Walk	All	100.0	999.0	9.1	4.7	0.0162
	Legs	100.0	999.0	9.7	4.3	0.0106
	Upper	0.0	204.0	22.8	9.3	0.0832
Dance	All	100.0	499.0	5.9	3.6	0.0090
	Legs	100.0	499.0	7.9	5.3	0.0101
	Upper	0.0	215.9	21.2	10.5	0.1413
Jump	All	100.0	499.0	46.1	24.2	0.0177
	Legs	100.0	499.0	41.9	22.4	0.0204
	Upper	95.8	487.6	71.5	36.1	0.0372
Loco.	All	99.2	995.7	—	—	0.0129
	Legs	99.3	997.3	—	—	0.0171
	Upper	91.0	964.1	—	—	0.0314

survival on the motion-tracking tasks and about 99% on locomotion. Upper-body inputs instead fail on walk and dance, and reconstruction RMSE is lower with leg inputs than upper-body inputs on every task. This supports the leg chain as the main source of the inertial information, though the upper body still contributes, most visibly on dance, where its arm and torso motion couples into the base. We retain all joints for consistent control across tasks and a uniform input configuration.

C. Comparison with Baselines

With the configuration fixed by the ablations, Table III(a) reports survival, episode length, and motion fidelity on the motion-tracking teachers, and Table III(b) reports survival and command tracking on the locomotion teacher. Figure 1 shows paired rollouts for each teacher.

JOSE and both distillation baselines achieve near-complete survival on the motion-tracking tasks, but their fidelity differs. Joint-only distillation has global position errors of 21.5, 13.6, and 94.2 mm on walk, dance, and jump, against 9.1, 5.9, and 46.1 mm for JOSE, less than half on every motion. On command-conditioned locomotion all deployable methods are within 0.015 of the teacher’s normalized tracking error, JOSE within 0.002, and the joint-only and IMU-based students differ by less than the spread across seeds.

The teacher driven by SET estimates reaches the lowest jump error (45.2 mm) but, trained offline as published,

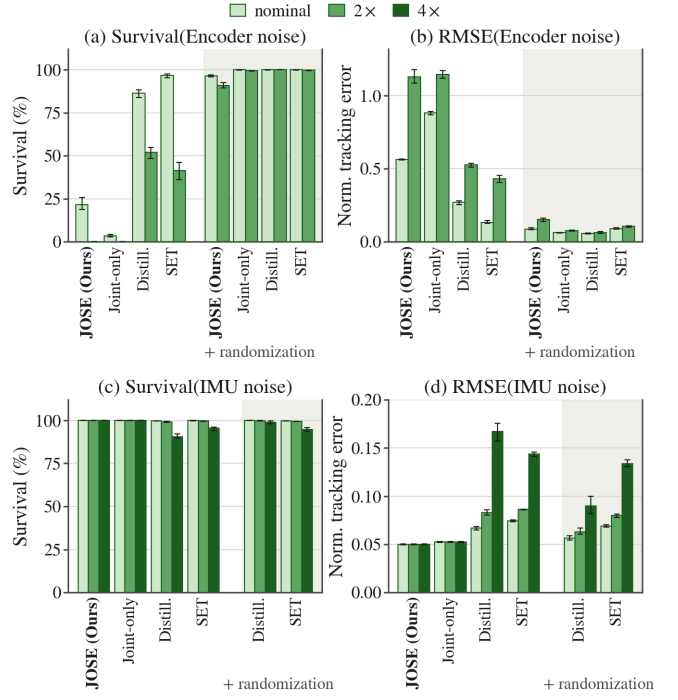


Fig. 5. **Sensor degradation on the locomotion task.** Panels (a, c) show survival and (b, d) normalized tracking error. The shaded block in each panel shows the same methods retrained with noise randomization for the corresponding sensor. The encoder axis is swept at nominal and $2\times$ noise, the IMU axis additionally at $4\times$.

fails on walk and dance after 145 and 164 steps on average. Table IV examines training effects on command-conditioned locomotion, where DAGger substantially reduces SET’s tracking-error gap to JOSE.

D. Sensor Degradation

We test robustness to sensor degradation against the same baselines (Fig. 5). Encoder noise is ± 0.01 rad in joint position and ± 1.5 rad/s in joint velocity, swept at nominal and twice nominal, and IMU noise is gyroscope white noise $\sigma = 0.015$ rad/s, per-episode gyroscope bias $\sigma = 0.01$ rad/s, and gravity tilt $\sigma = 0.015$ rad, swept at nominal, twice, and four times nominal, with an IMU input delay of 0 to 2 steps applied throughout. JOSE and the baselines are trained with and without sensor-noise randomization, the teacher frozen throughout, and IMU noise leaves JOSE and joint-only distillation unchanged because neither reads the IMU.

Without noise randomization during training (Fig. 5(a)), nominal encoder noise reduces JOSE survival to 21.5% and joint-only distillation to 3.5%, while the IMU-based methods survive. Training JOSE with encoder noise raises survival to 96.5% at nominal and 91.0% at twice nominal, while its clean normalized tracking error rises from 0.050 to 0.055 (Fig. 5(b)).

For IMU degradation, Fig. 5(c) reports survival and Fig. 5(d) normalized tracking error. At four times nominal noise, JOSE and joint-only distillation hold at their clean errors of 0.050 and 0.052, whereas noise-aware IMU-based distillation and SET reach 0.090 and 0.134 (Fig. 5(d)).

TABLE II
ESTIMATOR ARCHITECTURE ABLATION

(a) Motion-tracking teachers									
Task	Estimator	Surv. (%) $\uparrow$	$E_{g\text{-mpipe}} \downarrow$ (mm)	$E_{\text{mpipe}} \downarrow$ (mm)	Reconstruction RMSE $\downarrow$				
					Height (m)	Ori.	v (m/s)	ω (rad/s)	Bodies (m)
Walk	LSTM	100.0 $\pm$ 0.0	9.1 $\pm$ 1.9	4.7 $\pm$ 1.3	0.0004	0.0281	0.0274	0.0299	0.0068
	History MLP	9.7 $\pm$ 16.7	19.4 $\pm$ 13.9	10.4 $\pm$ 5.4	0.0045	0.0744	0.0905	0.1664	0.0188
	TCN	0.0 $\pm$ 0.0	24.8 $\pm$ 1.9	9.8 $\pm$ 1.5	0.0064	0.1165	0.1389	0.2147	0.0297
Dance	LSTM	100.0 $\pm$ 0.0	5.9 $\pm$ 2.0	3.6 $\pm$ 0.8	0.0006	0.0108	0.0082	0.0276	0.0027
	History MLP	0.0 $\pm$ 0.0	28.4 $\pm$ 4.5	14.1 $\pm$ 2.5	0.0124	0.0762	0.1232	0.2786	0.0259
	TCN	0.0 $\pm$ 0.0	38.7 $\pm$ 9.5	15.9 $\pm$ 2.5	0.0104	0.0923	0.1326	0.3158	0.0324
Jump	LSTM	100.0 $\pm$ 0.0	46.1 $\pm$ 6.4	24.2 $\pm$ 3.7	0.0035	0.0203	0.0248	0.0471	0.0087
	History MLP	76.3 $\pm$ 5.2	131.0 $\pm$ 15.2	58.8 $\pm$ 6.7	0.0188	0.0829	0.1312	0.2527	0.0356
	TCN	88.0 $\pm$ 4.8	161.0 $\pm$ 13.2	70.5 $\pm$ 1.6	0.0166	0.0747	0.1116	0.1962	0.0317

(b) Command-conditioned locomotion teacher							
Estimator	Surv. (%) $\uparrow$	Command tracking RMSE $\downarrow$			Reconstruction RMSE $\downarrow$		
		v_x (m/s)	v_y (m/s)	ω_z (rad/s)	v (m/s)	ω (rad/s)	g
LSTM	99.2 $\pm$ 0.6	0.024	0.037	0.053	0.0090	0.0202	0.0027
History MLP	96.5 $\pm$ 1.0	0.027	0.039	0.061	0.0194	0.0408	0.0059
TCN	91.2 $\pm$ 2.1	0.068	0.071	0.094	0.0277	0.0555	0.0079

TABLE III
DEPLOYMENT ACROSS ALL FOUR TEACHERS WITHOUT PRIVILEGED
OBSERVATION.

(a) Motion-tracking teachers					
Task	Method	Surv. (%) $\uparrow$	Ep. len. $\uparrow$	$E_{g\text{-mpipe}} \downarrow$	$E_{\text{mpipe}} \downarrow$
Walk	Teacher	100.0	999.0	0.0	0.0
	Distillation	99.7	997.7	11.4 $\pm$ 1.9	5.2 $\pm$ 0.9
	Joint-only distill.	100.0	999.0	21.5 $\pm$ 17.1	7.7 $\pm$ 2.9
	SET [4]	0.0	145.2	9.9 $\pm$ 3.9	5.6 $\pm$ 1.9
	JOSE (Ours)	100.0	999.0	9.1 $\pm$ 1.9	4.7 $\pm$ 1.3
Dance	Teacher	100.0	499.0	0.0	0.0
	Distillation	100.0	499.0	18.0 $\pm$ 1.9	6.7 $\pm$ 1.0
	Joint-only distill.	100.0	499.0	13.6 $\pm$ 5.1	8.4 $\pm$ 2.5
	SET	0.0	163.7	9.6 $\pm$ 4.0	5.4 $\pm$ 2.4
	JOSE (Ours)	100.0	499.0	5.9 $\pm$ 2.0	3.6 $\pm$ 0.8
Jump	Teacher	100.0	499.0	0.0	0.0
	Distillation	100.0	499.0	90.5 $\pm$ 7.1	45.6 $\pm$ 5.2
	Joint-only distill.	99.9	498.5	94.2 $\pm$ 22.9	47.1 $\pm$ 11.0
	SET	99.2	496.4	45.2 $\pm$ 4.3	23.2 $\pm$ 2.5
	JOSE (Ours)	100.0	499.0	46.1 $\pm$ 6.4	24.2 $\pm$ 3.7

(b) Command-conditioned locomotion teacher			
Method	Surv. (%) $\uparrow$	Norm. track err $\downarrow$	Δ teacher $\downarrow$
Teacher	100.0	0.048	—
Distillation	100.0	0.054	+0.005
Joint-only distill.	99.6	0.052	+0.004
SET [4]	100.0	0.063	+0.015
JOSE (Ours)	99.2	0.050	+0.002

E. Dataset Aggregation and IMU Dependence

JOSE is robust to IMU noise partly because it never reads the IMU, so this study asks whether reconstruction itself helps, beyond simply avoiding the sensor. We add JOSE+IMU, which reads a noisy IMU but still reconstructs all nine state components, and compare it with SET, which passes the six IMU-derived components through unchanged (Table IV). We also vary training, offline versus DAgger, with DR marking domain randomization of the IMU noise during

TABLE IV
DATASET AGGREGATION AND IMU DEPENDENCE ON
COMMAND-CONDITIONED LOCOMOTION.

Method	Reads	Training	Normalized tracking error $\downarrow$			
			Clean	Nom.	2 $\times$	4 $\times$
JOSE	Enc.	DAgger	0.0499	0.0499	0.0499	0.0499
JOSE+IMU	Enc.+IMU	DAgger	0.0481	0.0621	0.0851	0.2103
		DAgger+DR	0.0500	0.0500	0.0506	0.0532
SET	Enc.+IMU	Offline	0.0629	0.0743	0.0861	0.1438
SET+DAgger	Enc.+IMU	DAgger	0.0502	0.0625	0.0744	0.1265
		DAgger+DR	0.0504	0.0598	0.0705	0.1182

training. The IMU-noise settings match those in Fig. 5.

Effect of DAgger. DAgger reduces SET’s clean normalized tracking error from 0.0629 to 0.0502, close to JOSE’s 0.0499, so most of the clean-condition gap between them follows from training on DAgger rollouts rather than from the estimator architecture.

Effect of reconstructing the IMU. Adding the IMU to JOSE lowers its clean normalized tracking error to 0.0481, but under IMU noise its tracking performance deteriorates. Without DR, JOSE+IMU rises to 0.2103 at four times nominal noise. With DR, JOSE+IMU holds at 0.0532 while SET+DAgger reaches 0.1182 at the same noise, despite matched clean errors (0.0500 and 0.0504), so reconstructing the corrupted inertial state helps beyond randomization alone.

F. Terrain and Disturbance

We test locomotion beyond flat ground, under friction variation, on slopes, and under pushes, to find where encoder-only recovery holds and where it breaks (Fig. 6). Friction varies across environments in [0.3, 1.0], pyramid slopes reach gradients of 0.4, and pushes perturb base velocity by up to 0.5 m/s in x and y every 1 to 4 s. Each terrain condition uses its own teacher and deployment models, with IMU-noise

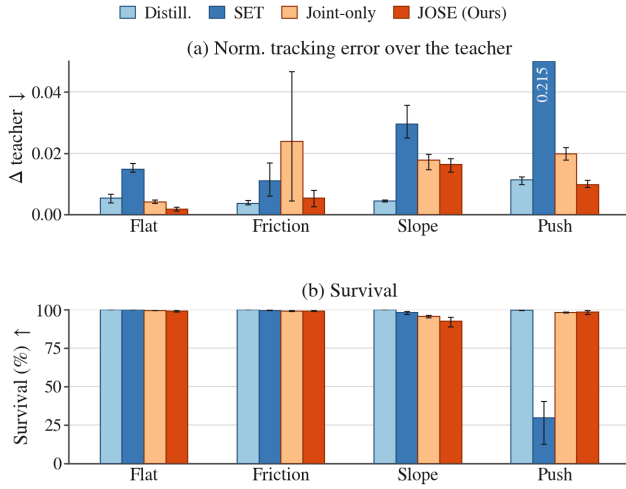


Fig. 6. **Locomotion under terrain variation and pushes.** Panel (a) shows normalized tracking error relative to each condition’s ground-truth-driven teacher, and (b) shows survival.

randomization for the IMU-based methods. On slopes we disable the terrain curriculum so each environment stays at a fixed steepness, and set the hardest level to the steepest slope the teacher itself survives, so any fall is the estimator’s doing rather than the teacher’s. The two joint-history methods, JOSE and joint-only distillation, did not converge on slopes at the budget used elsewhere, so both are trained there at twice that budget, 20 DAGGER rounds and 600 student iterations, and JOSE still trails on slopes even with this extra budget.

Friction and pushes. Under friction variation and pushes, JOSE’s normalized tracking-error increases over the corresponding teachers are 0.005 and 0.010, close to IMU-based distillation (0.004 and 0.011) and below joint-only distillation (0.024 and 0.020), at 99.2% and 98.5% survival (Fig. 6).

Slopes. On slopes JOSE reaches a normalized tracking-error increase of 0.016 at 92.7% survival, against 0.004 and 100% for IMU-based distillation (Fig. 6).

G. Yaw-Rate Tracking

Yaw rate is the axis least likely to be recoverable from encoders, since it is the rotation an IMU gyroscope measures directly, and failing it would leave the JOSE-driven teacher unable to make turning motions. The aggregate error in Table III(b) combines it with the linear axes, so we evaluate it on its own, over commands $\omega_z \in \{0, \pm 0.25, \pm 0.5, \pm 0.75, \pm 1.0\}$ rad/s with zero commanded linear velocity. Each command runs for 4 s after 1 s of settling, across 256 environments and three seeds, on flat ground with clean sensors.

Figure 7 reports command-tracking RMSE for yaw rate, with the ground-truth-driven teacher as a baseline. JOSE stays within 0.006 rad/s of the teacher across the commands, matching SET even though SET reads the gyroscope that measures angular velocity directly. Joint-only and IMU-based distillation exceed the teacher’s error by up to 0.014 and 0.019 rad/s.

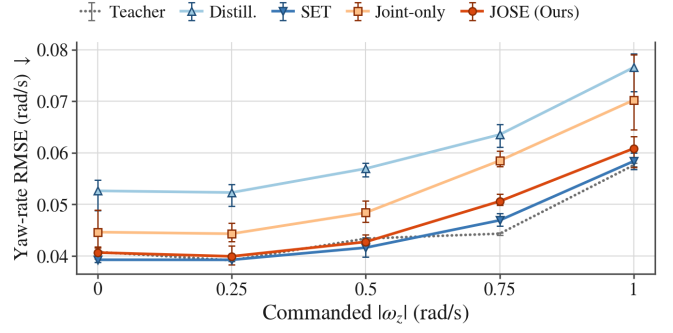


Fig. 7. **Yaw-rate tracking on flat ground with clean sensors.** Yaw-rate command-tracking RMSE is reported at each command magnitude after averaging over both turning directions. The dotted curve shows the ground-truth-driven teacher.

V. DISCUSSION

A. Why Survival Understates the Problem

Similar survival rates can hide differences in teacher fidelity. JOSE preserves the teacher’s motion better than either action-distillation baseline at comparable survival rates (Table III(a)). State reconstruction retains the teacher’s control map, whereas action distillation learns a new one. In locomotion, JOSE matches SET’s yaw-rate tracking despite using no gyroscope (Fig. 7). Stance-foot constraints may provide cues about base angular motion through leg kinematics even when absolute heading is undetermined. These results demonstrate accurate yaw-rate tracking without IMU inputs, but do not establish general observability from joint history.

B. The Cost and Benefit of Dropping the IMU

Reading only the encoders is a trade, not a free gain. It removes any sensitivity to IMU noise (Fig. 5) but concentrates the rest on the encoders, where noise-aware training becomes necessary. The benefit is also not merely from avoiding the IMU. Given a noisy IMU, JOSE reconstructs the inertial state rather than passing it through as SET does, and maintains low tracking error where the pass-through estimator degrades (Table IV).

C. When Joint-Only Estimation Holds

Leg-joint histories retain nearly all the performance of all-joint input, whereas upper-body histories do not (Table I). One explanation is stance kinematics: when the stance foot remains fixed in position and orientation, the leg chain constrains base pose and velocity relative to that foot [6], [7], whereas the upper body lies outside this chain. The ablation is consistent with this but does not isolate contact as the cause. The value of history is also task dependent, as at $W = 1$ jump retains 85.2% survival against 25.2% and 0% for walk and dance (Fig. 3(a)), arguing against bridging flight phases as its primary benefit.

D. Limitations

Each estimator is trained on one teacher’s distribution, and generalization across motions or teachers remains untested. Motion-tracking targets also include world-frame orientation, evaluated only within the training clips’ heading ranges.

Joint history does not determine base state independently of contact and terrain. Stance kinematics constrain motion relative to the surface but do not distinguish surface inclination from body tilt, as seen on slopes, where JOSE loses ground to the IMU-based baseline (Fig. 6). Slip and flight weaken the constraint further. The results therefore establish reconstruction within the evaluated tasks rather than general observability.

All experiments are in simulation. Sensor-noise evaluation does not establish robustness to real sensors or unmodeled dynamics, and physical deployment is unvalidated.

VI. CONCLUSION

JOSE reconstructs a frozen teacher’s privileged state, including the base orientation and angular velocity normally read from an IMU, from joint-encoder history alone. Across four simulated tasks it drives the teacher without an IMU, improving motion fidelity over joint-only distillation and matching the inertial baselines on locomotion. Reconstructing the inertial state rather than reading it also leaves control invariant to IMU noise. Encoder-noise sensitivity, the difficulty of inferring surface inclination, and dependence on the teacher’s distribution remain the main constraints, with physical deployment the natural next step.

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